

Machine Learning-Based Sentiment Analysis of Amazon Fine Food Reviews

Module: ITC 4378 - Big Data Analytics

Dataset: [Amazon Fine Food Reviews](#)

CPMs – 24422, 24414, 24379

Department of Information Technology
Faculty of Management Studies and Commerce
University of Sri Jayewardenepura

Introduction

Online customer reviews have become one of the most important sources of business intelligence in digital commerce. They influence purchase decisions, shape brand reputation, and provide direct evidence of customer satisfaction and dissatisfaction. Because the volume of review data is too large for manual inspection, sentiment analysis has become a core Natural Language Processing technique for automatically identifying opinion polarity and extracting actionable insights from textual feedback. In e-commerce settings, sentiment analysis supports product evaluation, customer experience monitoring, and recommendation improvement. [4]

This research focuses on sentiment analysis using the Amazon Fine Food Reviews dataset, a widely used benchmark corpus containing 568,454 reviews written by 256,059 users on 74,258 food products between October 1999 and October 2012. The dataset provides review text together with product, user, and rating information, making it suitable for both exploratory analysis and predictive modeling. Its scale and domain specificity make it appropriate for studying how customers express satisfaction and complaints about food products in online marketplaces. [2]

The broader research literature shows a clear methodological progression in review mining. Earlier work emphasized classical text features and machine learning, while recent studies increasingly compare recurrent neural networks, convolutional architectures, and transformer models such as BERT, RoBERTa, and XLNet. At the same time, the literature also shows that high classification accuracy alone is no longer enough. More recent studies emphasize aspect-based sentiment analysis, interpretability, domain adaptation, class imbalance, and the translation of model outputs into business decisions. These themes are directly relevant to the present project because a useful university study should not only label reviews as positive or negative, but also explain what customers care about and how those insights can be used to improve products and services. [5]

Literature review

Recent work in this area can be grouped into three broad strands. The first strand compares machine learning, deep learning, and transformer models on Amazon or e-commerce review datasets. In this strand, transformer models usually perform best. Ali, Hashmi, Yayilgan, and Shaikh reported that BERT outperformed the other evaluated approaches and reached 89% accuracy in Amazon product sentiment analysis. Pookduang and colleagues later found that RoBERTa outperformed Naive Bayes, KNN, CART, and LSTM on an Amazon book reviews dataset, reaching 96.30% accuracy and an F1-score of 98.11%. Daraghmi and Zyadeh's 2026 comparative study, which is especially relevant because it explicitly uses Amazon Fine Food Reviews alongside Amazon Unlocked Mobile Reviews, compared LR, SVM, CNN, and LSTM under a unified framework and highlights both class imbalance and the difficulty of reliably handling neutral sentiment. [7]

The second strand tries to make sentiment analysis more useful for product and platform decisions by moving beyond whole-review polarity. Davoodi, Mezei, and Heikkilä used aspect-based sentiment classification for e-commerce platform reviews, manually defining fourteen customer-satisfaction aspects and showing that RoBERTa achieved over 90% accuracy. Hashmi and Yayilgan extended standard review classification by adding product titles and headlines as context, comparing TF-IDF, Word2Vec, and FastText embeddings with multiple machine learning, deep learning, and transformer models. Their FastText-based stacked model achieved accuracy between 0.93 and 0.94 on several Amazon product categories, and the study also added LIME and topic modeling to improve interpretability. Vadla, Suresh, and Viswanathan similarly emphasized actionability; their study used BERT and T5 to detect aspects and emotions in Amazon reviews, and a later summary of that work reports accuracy around 92% for BERT and 91% for T5. [8]

The third strand consists of review and synthesis papers that explain where the field is heading and where it is still weak. Hua and colleagues' large systematic review of aspect-based sentiment analysis screened 727 primary studies and concluded that the literature suffers from lack of dataset and domain diversity, as well as domain mismatch between research settings and real applications. Daza and colleagues' 2024 bibliometric and systematic review of e-commerce product review sentiment analysis found that machine learning accounts for 48.1% of studies, deep learning for 44.4%, and hybrid methods for 7.4%, with Amazon and Twitter appearing as the most common data sources; the same review highlights SVM and LSTM as especially prominent. Alahmadi and coauthors' 2025 survey broadens the perspective further by arguing that the next major challenge is not simply benchmark accuracy, but generalization across domains, languages, and modalities, together with fairness and robustness. [9]

When the literature is read comparatively, a clear pattern emerges. The field has become very good at demonstrating that transformer models outperform many older baselines on benchmark datasets. However, fewer papers show how these models can be turned into interpretable, domain-specific decision tools. This matters for your assignment because a project on Amazon Fine Food Reviews should not stop at "which classifier is best." A stronger academic contribution is to ask which food-related issues drive negative sentiment, which complaints are most persistent or most helpful to other users, and how classification outputs can support practical recommendations. The most useful recent papers are exactly those that move in this direction: aspect-based analysis, context-aware learning, and explainability. [10]

Directly relevant food-domain studies are fewer, but they are still helpful. Sarraf’s 2024 study explicitly used an Amazon food reviews dataset, tested TF-IDF, Word2Vec, and Bag of Words features with Logistic Regression, Decision Tree, and Random Forest, and reported test accuracy between 85% and 89%, with Logistic Regression plus Bag of Words performing best at 89%. This is useful for your project because it shows that strong classical baselines remain competitive, especially for course projects where interpretability, runtime, and reproducibility also matter. [11]

Comparison table of recent studies

Authors	Year	Dataset	Methodology	Key findings	Limitations
Sarraf	2024	Amazon food reviews dataset	TF-IDF, Word2Vec, BoW with Logistic Regression, Decision Tree, Random Forest	Accuracy ranged from 85% to 89%; Logistic Regression with BoW performed best at 89%, which makes it a strong classical baseline for food-review sentiment work. [11]	Focuses on coarse review-level classification and classical ML only; no aspect-level modeling or transformer benchmark. [11]
Ali, Hashmi, Yayilgan, Shaikh	2024	Amazon product reviews	Comparative study of machine learning, deep learning, and transformer-based methods	BERT was reported as the best-performing model, reaching 89% accuracy. [12]	Broad Amazon setting rather than food-specific; mainly comparative classification, so actionability beyond polarity is limited. [12]
Vadla, Suresh, Viswana than	2024	Amazon reviews	BERT and T5 in a pipeline to detect aspects and customer emotions	The study emphasizes product-design support; later literature summarizes the model performance as about 92% for BERT and 91% for T5. [13]	Strongly design-oriented and not tied to the Fine Food corpus; relies on constructed aspect/emotion pipeline rather than plain review polarity alone. [13]
Daza, Gonzá le z Rueda, Aguilar	2024	Review paper over e-commer ce	Bibliometric analysis and systematic literature review	Found ML in 48.1% of studies, DL in 44.4%, hybrid in 7.4%; Amazon and Twitter	Review paper rather than a benchmark experiment;

Sánchez, Robles Espíritu, Chauca Quiñones		sentiment studies		were the dominant data sources, and SVM/LSTM were especially prominent. [14]	useful for mapping trends and gaps, not for selecting one “best” model directly. [15]
Hua, Denny, Wicker, Taskova	2024	727 ABSA studies across domains	Systematic literature review of aspect-based sentiment analysis	Identified systemic lack of dataset and domain diversity and domain mismatch in ABSA research. [16]	Not specific to Amazon food reviews; synthesizes the field rather than testing a concrete food-review model. [16]
Hashmi, Yayilgan	2024 online publication, 2025 volume	Four Amazon product-review datasets in communication technology domains	Product context-aware learning with product titles/headlines; TF-IDF, Word2Vec, FastText; XGBoost, CatBoost, SVM, LSTM, GRU, CNN, BERT, RoBERTa, XLNet; LIME and LDA for XAI	FastText with stacked XGBoost/CatBoost reached 0.93–0.94 accuracy on several datasets; FastText + linear SVM reached 0.91 on software reviews; the study also improved interpretability with LIME and topic modeling. [17]	Still limited to Amazon product-review settings and category-specific datasets; future work explicitly proposes going beyond one platform/type of review. [17]
Pookduang, Klangbunrueang, Chansanam, Lunrasri	2025	Amazon book reviews	Compared Naive Bayes, KNN, CART, LSTM, and RoBERTa	RoBERTa achieved 96.30% accuracy and 98.11% F1, outperforming both traditional ML and LSTM. [18]	Uses book reviews rather than food reviews; authors note computational-efficiency challenges and future multilingual/cross-domain work. [18]

Davoodi, Mezei, Heikkilä	2025 online publication, 2026 issue	Trustpilot reviews of five e-commerce platforms	Aspect-based sentiment classification on 14 manually defined aspects using recent deep learning models	RoBERTa achieved over 90% accuracy; the study shows that aspect-level sentiment can reveal concrete drivers of customer satisfaction. [19]	Focuses on platform satisfaction, not food products; requires manual aspect design and annotation effort. [19]
Alahmadi, Alharbi, Chen, Wang	2025	Review across sentiment-analysis domains	Survey of progress, challenges, and emerging directions	Highlights domain adaptation, linguistic diversity, multimodal integration, self-supervised learning, fairness, and robustness as next-stage challenges. [20]	Broad survey rather than Amazon-specific experimentation. [20]
Daraghmi, Zyadeh	2026	Amazon Fine Food Reviews and Amazon Unlocked Mobile Reviews	Unified comparison of LR, SVM, CNN, and LSTM with preprocessing that explicitly considers negation and sarcasm	Directly relevant to your dataset; the study foregrounds label imbalance, compares ML and DL under the same framework, and notes that CNN was fastest but more prone to overfitting and weaker on neutral sentiment. [21]	Uses coarse review-level sentiment classification on two datasets; despite better preprocessing, neutral handling and imbalance remain challenging. [22]

Research gap

The clearest gap is food-domain specificity. Many recent Amazon studies focus on books, electronics, software, communication technology products, or generic platform reviews, while relatively few recent papers directly target the Amazon Fine Food Reviews corpus. That makes a food-specific sentiment project academically justifiable, especially if it surfaces domain-specific themes such as taste, quality, packaging, freshness, ingredient concerns, or value for money. [23]

A second gap is the dominance of coarse review-level polarity over aspect-level insight. The strongest recent methodological papers show that aspect-based approaches are more actionable than single-label sentiment classification, yet many Amazon review papers still benchmark only overall polarity or rating classes. For a Big Data Analytics project, this is important because prescriptive analytics depends on knowing *what* is wrong, not only *whether* a review is negative. [24]

A third gap is generalization and domain mismatch. Hua and colleagues show that ABSA research still suffers from limited domain diversity and domain mismatch, while Alahmadi and colleagues argue that cross-domain adaptability remains a central unsolved problem. A model trained on one review domain may not generalize well to another, and this matters for your dataset because Amazon Fine Food Reviews is large but also old and domain-specific. [26]

Motivation to the project

This project is motivated by the growing number of customer reviews that businesses cannot analyze manually. It will use sentiment analysis to classify reviews and compare different models. The findings will help identify common customer concerns and positive experiences, allowing sellers and product managers to make better decisions, improve products, and increase customer satisfaction.

References

[1] [4] [20] Generalizing sentiment analysis: a review of progress, challenges, and emerging directions | Social Network Analysis and Mining | Springer Nature Link

<https://link.springer.com/article/10.1007/s13278-025-01461-8>

[2] [29] [32] Web data: Amazon Fine Foods reviews - SNAP: Stanford

<https://snap.stanford.edu/data/web-FineFoods.html>

[3] [8] [19] [24] [28] Aspect-based sentiment classification of user reviews to understand customer satisfaction of e-commerce platforms | Electronic Commerce Research | Springer Nature Link

<https://link.springer.com/article/10.1007/s10660-025-09948-4>

[5] [17] [25] [34] A robust hybrid approach with product context-aware learning and explainable AI for sentiment analysis in Amazon user reviews | Electronic Commerce Research | Springer Nature Link

<https://link.springer.com/article/10.1007/s10660-024-09896-5>

[6] [11] [30] [31] [33] Utilizing NLP Sentiment Analysis Approach to Categorize Amazon Reviews against an Extended Testing Set | International Journal of Computer (IJC)

<https://ijcjournal.org/InternationalJournalOfComputer/article/view/2199>

[7] [12] Analyzing Amazon Products Sentiment: A Comparative ...

<https://www.mdpi.com/2079-9292/13/7/1305>

[9] [16] [26] A systematic review of aspect-based sentiment analysis: domains, methods, and trends | Artificial Intelligence Review | Springer Nature Link

<https://link.springer.com/article/10.1007/s10462-024-10906-z>

[10] [18] [23] Advancing Sentiment Analysis: Evaluating RoBERTa against Traditional and Deep Learning Models | Engineering, Technology & Applied Science Research

<https://etasr.com/index.php/ETASR/article/view/9703>

[13] "Enhancing Product Design through AI-Driven Sentiment ...

https://scholarworks.sjsu.edu/faculty_rsca/4542/

[14] [15] Sentiment Analysis on E-Commerce Product Reviews ...

<https://www.sciencedirect.com/science/article/pii/S2667096824000569>

[21] Sentiment Classification of Amazon Product Reviews ...

https://www.researchgate.net/publication/401652097_Sentiment_Classification_of_Amazon_Product_Reviews_Based_on_Machine_and_Deep_Learning_Techniques_A_Comparative_Study

[22] [27] Sentiment Classification of Amazon Product Reviews ...

<https://www.mdpi.com/1999-5903/18/3/138>

Dataset: <https://www.sciencedirect.com/science/article/pii/S2667096824000569>